Kibana 9.5 Cheat Sheet

Dashboards, alerting and as code. Lab: Docker Compose kibana profile (EDOT Collector, Elasticsearch, Kibana).

STACK AND INGEST CHOICES

- Elasticsearch 9.5.6 and Kibana 9.5.6 (same version for every component); ports `9200` and `5601`.

EDOT / OpenTelemetry	default for new work; EDOT Collector is built into Elastic Agent from 9.5
Agent + Fleet	managed integrations with dashboards and pipelines
Beats	existing estates only
Logstash	heavy transformation or fan-out
APM Server	classic Elastic APM agents

DATA STREAMS AND TSDS

Naming `<type>-<dataset>-<namespace>`.

```
logs-*-* metrics-*-* traces-*-*
logs-*.otel-* metrics-*.otel-*
traces-*.otel-* (EDOT, OTLP)
metrics-<dataset>.prometheus-default
  (Prometheus remote write)
```

- OTel default dataset `generic`, namespace `default`.
- TSDS for metrics; a custom template overlapping `metrics-*-*` needs priority above 500.
- Lifecycle: ILM or data stream lifecycle sets retention.

KIBANA.YML MUST-HAVES

```
xpack.encryptedSavedObjects.
  encryptionKey: <32+ chars>
server.publicBaseUrl:
  "https://kibana.example.com"
elasticsearch.hosts: [...]
elasticsearch.serviceAccountToken
```

- Without the encryption key alerting and actions are blocked; without `publicBaseUrl` notification links break.
- Keep secrets in `kibana-keystore`, not in the file.
- Key rotation: `decryptionOnlyKeys`.

KQL, ES|QL, PROMQL

KQL filters documents (bar `not` > `and` > `or`):

```
service.name : "bookshelf" and
  severity_text : "ERROR"
```

ES|QL aggregates (`FROM`, `TS`, `PROMQL`):

```
FROM logs-*.otel-*
| STATS errors = COUNT(*)
    WHERE severity_text == "ERROR",
    total = COUNT(*)
  BY b = BUCKET(@timestamp, 50,
    ?_tstart, ?_tend)
| EVAL share = TO_DOUBLE(errors)/total
```

PromQL via ES|QL:

```
PROMQL index=metrics-*
  r=(sum(rate(m[5m])))
```

- `COUNT` is an integer: convert before dividing. `TS` only for time series data streams.

LENS CHART PICKER

Trend	Line, area; bar for buckets
Now	Metric (with trend); Gauge
Distribution	Bar with histogram; Heat map
Parts	Pie, donut, treemap
Top N	Table
Where	Maps
Logs, traces	Discover session, APM views

Formula, a ratio:

```
count(kql='http.response.status_code >= 500')
  / count()
```

- Empty buckets give no point, not zero. Also: time shift, reference lines, annotations.

CONTROLS AND DRILLDOWNS

- Controls: options list, range slider, time slider (needs a time field); pin to scope them.
- Only an ES|QL variable control substitutes into a query (`??field`, `?value`); others add filters.
- Drilldowns: dashboard, URL, Discover.
- Dashboards as code: `/api/dashboards` (GET, POST, PUT by id, DELETE); PUT replaces the whole dashboard.

THE THREE ALERTING SYSTEMS

Classic Alerting	GA; form-based rules, connectors, actions
Universal Alerting	experimental in 9.5, GA documented from 9.6; event history
Watcher	self-managed only; scripting, chained inputs

- Rule = check interval + conditions + alert delay (consecutive runs) + actions.

RULE TYPES

Stack	`.es-query` (query, ES|QL), `.index-threshold`, `.geo-containment`, `transform_health`
Custom threshold	`observability.rules.custom_threshold`; groups `fired`, `warning`, `nodata`
APM	`apm.transaction_duration`, `apm.error_rate`, `apm.transaction_error_rate`, `apm.anomaly`
Logs, infra	`logs.alert.document.count`, `metrics.alert.inventory.threshold`
SLO burn rate	`slo.rules.burnRate`; groups `alert`, `high`, `medium`, `low`
Deprecated	metric threshold (9.4); use custom threshold

ACTION FREQUENCY

On check intervals	`onActiveAlert`: every check while active
Custom interval	`onThrottleInterval` + `throttle: "5m"` (not below the check interval)
On status changes	`onActionGroupChange`: once per change; recovery counts

- Per-alert or summary (`summary: true`); `alerts_filter` limits by query or timeframe.
- No pending state; a resolved message needs an action for the `recovered` group.
- Set frequency on actions, not also on the rule.

MUSTACHE VARIABLES

```
{{rule.name}} {{rule.id}} {{rule.url}}
{{alert.id}} {{alert.actionGroup}}
{{alert.flapping}} {{date}}
{{context.reason}} {{context.value}}
{{context.alertDetailsUrl}}
{{kibanaBaseUrl}}
```

- Triple braces skip escaping: `{{{x}}}`; double braces keep JSON valid.
- `{{{.}}}` as a webhook body shows every variable the rule provides.
- Lambdas: `{{#FormatDate}}`, `{{#EvalMath}}`, `{{#ParseHjson}}`; `asJSON` suffix.

CONNECTOR TYPES

Stack	Index, Server log, Cases
Alerting	Email, Slack, Teams, PagerDuty, Opsgenie, Jira, ServiceNow, Webhook
Security, AI	CrowdStrike, SentinelOne, Defender; OpenAI, Bedrock, Gemini deprecated in 9.5

- `xpack.actions.allowedHosts` defaults to `["*"]`; restrict it.

ALERT STATUSES, FLAPPING, SNOOZE

active	conditions met; actions per frequency
flapping	switches often; status-change actions stop
recovered	conditions unmet after being met; final
untracked	rule disabled or marked untracked

- Flapping: `flapping.enabled`, `look_back_window`, `status_change_threshold`; leave it after that many quiet runs.
- Snooze keeps rules running: `POST /api/alerting/rule/{id}/snooze_schedule`; one alert: `.../alert/{alert_id}/_snooze`.
- Maintenance windows: `POST /api/maintenance_window`.

KEY APIS

```
POST /api/alerting/rule/{id}
GET /api/alerting/rules/_find
POST /api/actions/connector/{id}
POST /api/maintenance_window
GET /api/alerting/_health
GET /api/task_manager/_health
PUT /s/{space}/api/dashboards/{id}
```

- Writes need header `kbn-xsrf: true`; spaces use `/s/{space}/` prefix.
- UI says connectors, API says actions.

TERRAFORM (ELASTICSTACK)

```
elasticstack_kibana_alerting_rule
elasticstack_kibana_action_connector
elasticstack_kibana_maintenance_window
elasticstack_kibana_slo
elasticstack_kibana_data_view
elasticstack_kibana_dashboard (preview)
```

TASK MANAGER HEALTH

- `GET /api/alerting/_health` returns `decryption_health`, `execution_health`, `read_health` (`ok`, `warn`, `error`).
- Decryption errors usually mean a different encryption key.
- Tasks live in `.kibana_task_manager`; scale with `xpack.task_manager.capacity`.

REMOVED / DEPRECATED

`/api/alerts/*`	removed in 9.0; use `/api/alerting/rule(s)`
Logs Explorer, Logs Stream	removed in 9.0; Discover
Uptime	deprecated 8.15; Synthetics
Metric threshold rule	custom threshold
X-Pack, Shield, Marvel	built in; Stack Monitoring, AutoOps
Per-Beat install	Elastic Agent, EDOT